

上海交通大学试卷 (B 卷)

(2022 至 2023 学年 第 1 学期)

班级号 _____ 学号 _____ 姓名 _____

课程名称 _____ CS2601 线性优化与凸优化 _____ 成绩 _____

There are 7 problems and 100 points in total.

1. Determine whether the following functions are convex in their domain, and give your reasons.

(a) (3 points) $f(x, y) = x^2/y$, where $y > 0$.

(b) (3 points) $f(x) = |x^3|$, where $x \in \mathbb{R}$.

(c) (3 points) $f(x) = -x \log x - (1-x) \log(1-x)$, where $x \in (0, 1)$.

(d) (3 points) $f(x_1, x_2, x_3) = \max\{x_1^2 + x_2 - x_3, e^{x_2} + x_3^2 + 3x_1\} - \min\{x_1 + 2x_2 - 3x_3, 5x_3 + 4x_2\}$, where $x_1, x_2, x_3 \in \mathbb{R}$.

(e) (3 points) $f(\mathbf{x}) = e^{\|A\mathbf{x} + \mathbf{b}\|_2}$, where $\mathbf{x} \in \mathbb{R}^n$.

2. (10 points) Show that the ellipsoid

$$E = \left\{ (x, y, z) : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \right\}$$

is a convex set, where $a, b, c \in \mathbb{R}$ are three constants.

3. Consider the following optimization problem:

$$\begin{aligned} \text{minimize} \quad & f(x) = \begin{cases} x^2, & x \leq 1; \\ 2 - x, & x \geq 1; \end{cases} \\ \text{subject to} \quad & x + 1 \leq 0. \end{aligned}$$

(a) (2 points) What is the optimal solution and optimal value?

(b) (6 points) Find the dual problem.

(c) (2 points) Does strong duality hold?

4. Consider the following optimization problem,

$$\begin{aligned} \text{minimize} \quad & f(x_1, x_2) = e^{x_1 + x_2} - x_1 \\ \text{subject to} \quad & x_1 + 2x_2 = 1 \end{aligned} \tag{1}$$

(a) (2 points) Write down an equivalent unconstrained problem by eliminating x_2 .

(b) (3 points) For the unconstrained problem you find in (a), find the next iterate $x_1^{(1)}$ given by gradient descent with initial point $x_1^{(0)} = -1$ and step size $t = 1$. Also find the corresponding $x_2^{(1)}$ so that $(x_1^{(1)}, x_2^{(1)})$ is feasible.

(c) (5 points) Now we use *Newton's method with equality constraints* to solve the original problem (1). Write down the KKT system at $\mathbf{x}^{(0)} = (-1, 1)$, and find the next iterate $\mathbf{x}^{(1)}$ using step size $t = 1$.

5. Consider the following primal optimization problem:

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|_2^2 \\ \text{subject to} \quad & \mathbf{a}^T \mathbf{x} \leq b \end{aligned}$$

where $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a} \neq \mathbf{0}$, $b \in \mathbb{R}$, and $\|\cdot\|_2$ denotes the Euclidean norm.

- (3 points) Is it a convex optimization problem? Show your argument.
- (5 points) Write down the KKT conditions.
- (5 points) Find the optimal value, the optimal solution, and the corresponding Lagrange multipliers.
- (5 points) Write down the dual problem, find the dual optimal solution and dual optimal value.
- (7 points) If we redefine the function $f(\mathbf{x})$ in the primal optimization problem as $f(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|_2$, what are the optimal value, the optimal solution, and the corresponding Lagrange multipliers? Here we only consider the case $b < 0$.

6. In this problem, we use the so-called *mirror descent* algorithm to solve the following constrained minimization problem,

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & \mathbf{x} \in C, \end{aligned}$$

where C is a convex set. Mirror descent is an iterative algorithm with the following update rule,

$$\mathbf{x}^{(k+1)} = \arg \min_{\mathbf{x} \in C} \left\{ \nabla f(\mathbf{x}^{(k)})^T (\mathbf{x} - \mathbf{x}^{(k)}) + \frac{1}{t_k} D_\varphi(\mathbf{x}, \mathbf{x}^{(k)}) \right\}, \quad (2)$$

where $\mathbf{x}^{(k)}$ is the k -th iterate, $t_k > 0$ is the step size, and D_φ is the so-called *Bregman divergence*. For a given strictly convex function φ on C , D_φ is defined by

$$D_\varphi(\mathbf{x}, \mathbf{y}) = \varphi(\mathbf{x}) - \varphi(\mathbf{y}) - \nabla \varphi(\mathbf{y})^T (\mathbf{x} - \mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C.$$

- (2 points) Show that $D_\varphi(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in C$.
- (3 points) The function φ is typically chosen to match the geometry of C . For (b)-(d), we consider the case that $C = \Delta = \{\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1, \mathbf{x} \geq \mathbf{0}\}$ is the probability simplex. In this case, we can choose the negative entropy function, i.e.,

$$\varphi(\mathbf{x}) = \sum_{i=1}^n x_i \log x_i.$$

Show that for this choice of φ , D_φ is the KL divergence for $\mathbf{x}, \mathbf{y} \in \Delta$, i.e.,

$$D_\varphi(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n x_i \log \frac{x_i}{y_i}$$

- (4 points) If the initial point $\mathbf{x}^{(0)}$ is strictly feasible, i.e. $\mathbf{x}^{(0)} > \mathbf{0}$ and $\mathbf{x}^{(0)} \in \Delta$, it can be shown that all the iterates $\mathbf{x}^{(k)}$ will be strictly feasible. It turns out that the iterate $\mathbf{x}^{(k+1)}$ is the solution to the following optimization problem,

$$\begin{aligned} \text{minimize} \quad & F_k(\mathbf{x}) \triangleq \nabla f(\mathbf{x}^{(k)})^T (\mathbf{x} - \mathbf{x}^{(k)}) + \frac{1}{t_k} \sum_{i=1}^n x_i \log \frac{x_i}{x_i^{(k)}} \\ \text{subject to} \quad & \sum_{i=1}^n x_i = 1 \end{aligned}$$

Is this a convex problem? Explain your answer.

- (6 points) Find the closed-form update rule by solving the problem in (c).

7. The *Projected Gradient Descent* (PGD) is a simple way to solve constrained optimization problems. Consider the optimization problem

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g_1(\mathbf{x}) = 0, \dots, g_m(\mathbf{x}) = 0; \\ & h_1(\mathbf{x}) \leq 0, \dots, h_\ell(\mathbf{x}) \leq 0. \end{aligned}$$

Let $\Omega = \{\mathbf{x} : g_1(\mathbf{x}) = 0, \dots, g_m(\mathbf{x}) = 0, h_1(\mathbf{x}) \leq 0, \dots, h_\ell(\mathbf{x}) \leq 0\}$ be the feasible set.

Starting from an initial point $\mathbf{x}^{(0)} \in \mathbb{R}^n$, PGD iterates the following process until a stopping condition is met.

- Let $\mathbf{y}^{(k+1)} = \mathbf{x}^{(k)} - t_k \cdot \nabla f(\mathbf{x}^{(k)})$, where t_k is the step size. Namely, $\mathbf{y}^{(k+1)}$ is the next step in gradient descent if there are no constraints.
- Let $\mathbf{x}^{(k+1)}$ be the nearest point to $\mathbf{y}^{(k+1)}$ in Ω . Namely,

$$\mathbf{x}^{(k+1)} = \arg \min \|\mathbf{x} - \mathbf{y}^{(k+1)}\|_2^2$$

$$\min_{x \in \Omega} \|x - y\|_2$$

This step is called *projection*, denoted by $x^{(k+1)} = \mathcal{P}_\Omega(y^{(k+1)})$.

Now we would like to use PGD to solve the following problem:

$$\begin{aligned} &\text{minimize} && 5e^{x_1+x_2} + x_2^2 - 6x_1 \\ &\text{subject to} && x_1 + x_2^2 \leq 0 \end{aligned}$$

- (a) (3 points) Sketch the feasible set Ω .
- (b) (3 points) Suppose we start from $x^{(0)} = \mathbf{0}$, i.e. $x_1^{(0)} = 0$ and $x_2^{(0)} = 0$. What is $y^{(1)}$ if we set the step size $t_0 = 1$?
- (c) (4 points) Now we would like to compute $x^{(1)}$. Namely, we would like to solve the following optimization:

$$\begin{aligned} &\text{minimize} && (x_1 - y_1^{(1)})^2 + (x_2 - y_2^{(1)})^2 \\ &\text{subject to} && x_1 + x_2^2 \leq 0 \end{aligned}$$

Write down the KKT condition.

- (d) (5 points) Starting from $x^{(0)}$ in (b), find $x^{(1)}$ by solving the KKT condition.